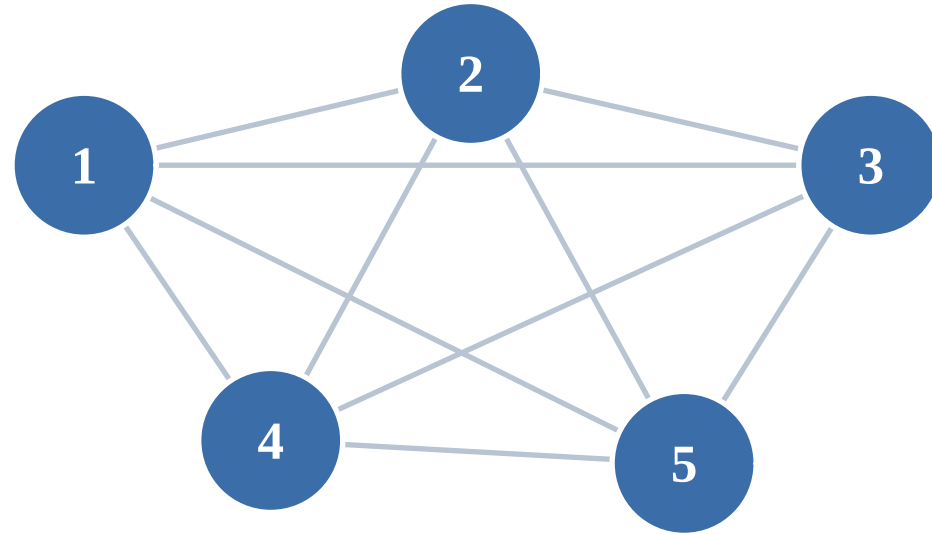


Attention and selective state-space information paths

The operators differ in what they compare directly and in the meaning assigned to sequence order

Multi-head attention

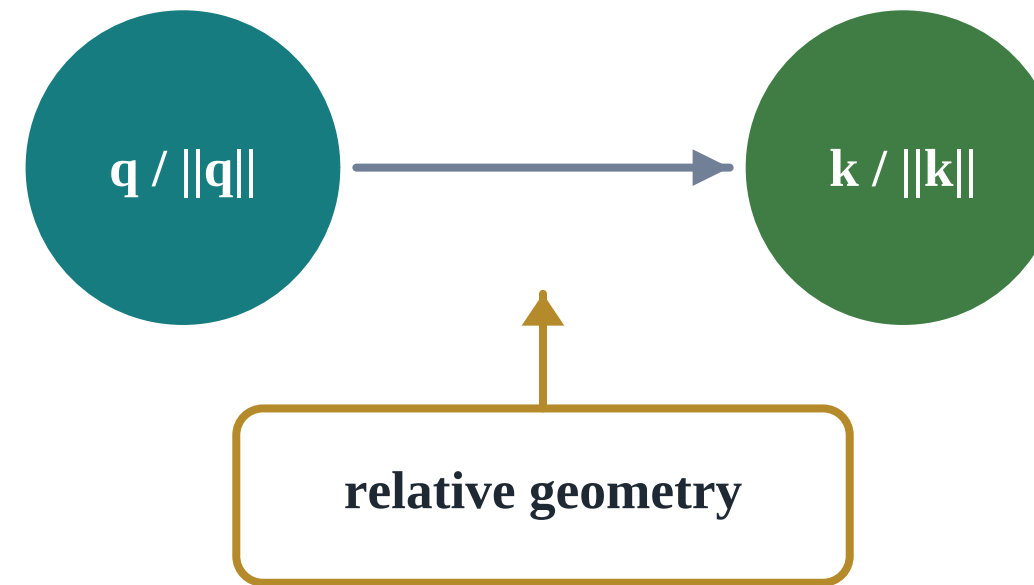


explicit all-to-all retrieval

content score: $q \cdot k$

sequence cost: $O(L^2)$

Normalised, relation-aware attention

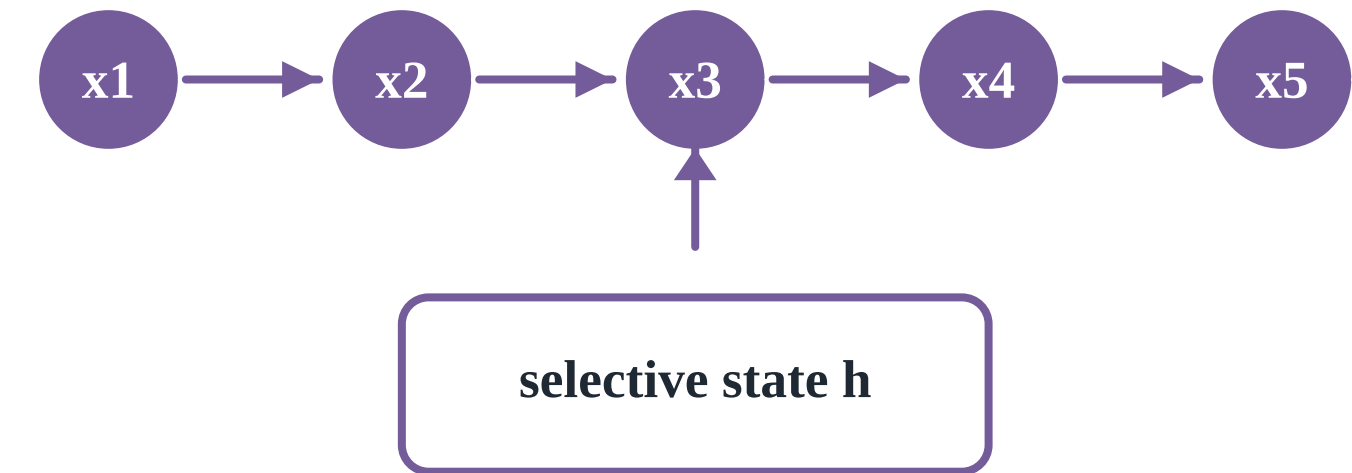


scaled cosine similarity

+ pairwise physical bias

sequence cost: $O(L^2)$

Selective SSM / Mamba



ordered recurrent summary

input-dependent retain / forget

sequence cost: $O(L)$

Hybrid design principle

Use attention for explicit interaction and selective recurrence for a genuinely ordered temporal axis